

9.1: Differentiating Exps. & logs.

$y = e^{f(x)}$

$f(x) = u \Rightarrow \frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = e^u \times f'(x) = f'(x)e^{f(x)}$

$y = \ln[f(x)]$

$\Rightarrow e^y = f(x)$

$\Rightarrow e^y \frac{dy}{dx} = f'(x) \Rightarrow \frac{dy}{dx} = \frac{f'(x)}{e^y} = \frac{f'(x)}{f(x)}$

$y = a^{f(x)} = e^{f(x)\ln a}$

$f(x)\ln a = u$

$\Rightarrow \frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = e^u \times f'(x)\ln a$

$= f'(x)a^{f(x)}\ln a$

2. Differentiating Trig Functions

$y = \sin(f(x))$, $u = f(x)$

$\Rightarrow \frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = f'(x)\cos(f(x))$

$y = \cos(f(x))$, $u = f(x)$

$\Rightarrow \frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = -f'(x)\sin(f(x))$

$y = \tan(f(x))$

$\Rightarrow \frac{d}{dx}\left(\frac{\sin(f(x))}{\cos(f(x))}\right) = \frac{f'(x)\cos^2(f(x)) + f'(x)\sin^2(f(x))}{\cos^2(f(x))} = f'(x)[\sec^2(f(x))]$

$= f'(x)\sec^2(f(x))$

3: Implicit Differentiation

An implicit function is not in form: $y = f(x)$. ID uses chain rule.

$\Rightarrow \frac{d}{dx}(f(y))$, let $f(y) = u \Rightarrow \frac{d}{dx}(u) = \frac{du}{dy} \times \frac{dy}{dx} = \frac{d}{dy}(f(y))\frac{dy}{dx}$

$\Rightarrow F(x, y) = 0$

$\Rightarrow \frac{dF}{dx} = \frac{\partial F}{\partial x} + \frac{\partial F}{\partial y}\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{-\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial y}} = \frac{-F_x}{F_y}$